

FR-380: Automated Testing

I. Functional Requirements

ID	Requirement	Description
FR-381	Synthetic Profile Management	A synthetic profile must include: • Profile name • Basic farm information • Symptoms • Picture (optional) • Label: the target disease Admin must be able to view/create/edit/delete profiles. Admin must be able to: • filter by name/label • search by name/id. • sort by name/label Admin must be able to do bulk action. Admin must be able to view list of profile with brief description of them: • Id • Name • Label
FR-382	Automated Test Execution	A testing session should have the following settings: Pre-run: • Pick tested model • Synthetic Profile Selection • Number of repetitions • (Optional) AI anomaly detection • (Optional) AI accuracy report The result of the test should include: • IDs of the generated chat • Report of returned errors (with link to them)

		• (Optional) Anomaly Report • (Optional) Accuracy Report
FR-383	AI Conversation Engine	LLM will be used to answer the chatbot questions using information from: • Synthetic Profile • Dictionary • Chat History
FR-384	Test Suite Management	Admin manages test suites with CRUD operations. Admin creates or updates a test suite by configuring the following settings for a test suite: • Which model • Which mode • What Synthetic Profile (SP) to use (allow preview). • Number of workers (default 1)
FR-385	Admin views test suite results	Upon execution, the system must return a list of test cases included, presenting the following metrics for each case: • Disease • Flu (Fluency) • Eff (Efficiency) • Lat (Latency) • Turns • Acc (Accuracy) • Cost • Step
FR-386	Admin views full conversation details	Admin can select a specific test case to "Show full conversation". The detailed view must include: • Chat history (SP vs Chatbot). • Evolution/history of hypothesis (at every step). • Reasoning. • Scores (Accuracy, Fluency, etc.). • Lat breakdown table.

		• Chatbot's feedback.
FR-387	Admin downloads reports	Download • Report as .md • Raw json data Summary Report: • See doc farm2vets.com/app/chatbot/reports/eval-en • Download ◦ Report as .md ◦ Raw json data

II. Data schema validation

```

1  const TestCaseMetrics = z.object({
2  })
3
4  const ConversationDetails = z.object({
5  })
6
7  const TestCase = z.object({
8    testCaseId: z.string(),
9    syntheticProfileId: z.string(),
10   metrics: TestCaseMetrics,
11   conversation: ConversationDetails.optional(),
12 })
13
14 export const TestSuite = z.object({
15   testSuiteId: ObjectIdAsHexString,
16   model: z.string(),
17   mode: z.string(),
18   syntheticProfiles: z.array(z.string()),
19   numWorkers: z.number().default(1),
20   testCases: z.array(TestCase).optional(),
21   createdAt: UnixTimestamp,
22   updatedAt: UnixTimestamp.nullish(),
23 })

```

III. Use-case Specification

UC-381. Synthetic profile management

Summary: This UC describe the process where the user admin can view and edit the current existing synthetic profiles.

Goal: To provide an interface for the admins to manage the synthetic profiles for automated testing purposes.

Preconditions:

- The user has the admin rights.
- The user is currently in Admin interface. Admin

Post-conditions:

- Success: The user can view all the synthetic profiles and edit those profiles
- Failure: The user is unable to view the synthetic profiles and/or unable to edit them due to network errors or insufficient permissions

Main Success Scenario (Basic Flow)

This scenario mainly describe what user might see when they are viewing the list of questions

Step	Actor Action (User)	System Action (Frontend)
1	The user enters the Voew Profiles interface.	GOTO: /admin/profiles If the user has admin rights and fetch successfully: • Display the list of profiles, limit and offset by pages. ◦ The profiles in the list should at least contain the metadata: ▪ Name ▪ Disease Label • Options for filter also appear: ◦ By name/label (disease) • Options for sort also appear: ◦ By name/label • Options for searching also appear: ◦ By name/disease Else: • Return a screen which notifies the user that they are unauthorized.
2	The user filters, sorts and	Show the list of profiles according to the search parameters

	searches according to their preference.	
3a	The user clicks View button or Delete button	If View button is clicked: • GOTO <code>/admin/profiles/{profileId}</code> • Shows the information that is in the profile with editor available If Delete button is clicked: • Show a prompt to confirm the deletion action • If the user agrees, then DELETE the profile and return back to the user interface.

UC-382. Automated Test Execution

Summary: This UC describe the process that the admin can execute automated test execution for the user profiles.

Goal: To provide an interface for the admins to execute automatic test execution for testing the accuracy of the chatbot without having to manually entering the information to the chatbot.

Preconditions:

- The user has the admin rights.
- The user is currently in **Admin** interface.

1

Postconditions:

- Success: The user can perform automated test execution and receives the results of the tests performed
 - Failure: The user is unable to perform automated test execution due to network errors or insufficient permissions
-

Main Success Scenario (Basic Flow)

This scenario mainly describe what user might see when they are giving feedback.

This scenario mainly describe what user might see when they are querying the feedbacks from the database. We start from the

Step	Actor Action (User)	System Action (Frontend)
1	The user clicks the Execute Tests interface.	GOTO: /admin/tests If the user has admin rights and fetches successfully: • Display the interface to adjust parameters for tests: ◦ Choose synthetic profiles ◦ Choose number of repetitions Else: • Return a screen which notifies the user that they are unauthorized.
2	The user clicks Execute after filling in all the fields	If the user has admin rights and fetches successfully: • Starting to perform tests (Results can be updated in real-time) (Optional) • After the tests have been successfully performed, the user can receive all the data about the tests, including the IDs, the content and link to the tests Else: • Return a screen which notifies the user that they are unauthorized, or there are networks error.

UC-384. Test Suite Management

Summary: This UC describes the process where the Admin creates, reads, updates, or deletes test suites.

Goal: To help admin define parameters and manage collections of automated tests.

Preconditions:

- The user has the admin rights.
- The user is currently in Admin interface, Automated Testing Console.

Postconditions:

- Success: The test suite configuration is successfully saved, updated, deleted, or displayed.
- Failure: The system displays a validation or server error message.

Main Success Scenario (Basic Flow)

Step	Actor Action (Admin)	System Action (Frontend)
1	Admin clicks "Create Test" or selects an existing test to update.	GOTO: /link/to/ If the user has admin rights and fetches successfully: • Display the interface to adjust parameters for tests: ◦ Generative Model ◦ Mode ◦ Choose synthetic profiles ◦ Choose number of repetitions
2	Admin clicks "Save".	System persists the test suite document and displays a success message.

UC-385. View Test Suite Result

Summary: This UC describes the process where the Admin views the results of an executed test suite.

Goal: Allow the Admin to see an overview of all run test cases and their performance metrics.

Preconditions:

- The user has the admin rights.
- The user is currently in Admin interface, Automated Testing Console.
- The test suite has been successfully executed.

Postconditions:

- Success: The system returns and displays a list of all included test cases.
- Failure: The system displays a validation or server error message.

Main Success Scenario (Basic Flow)

Step	Actor Action (Admin)	System Action (Frontend)
1	Admin navigates to a completed Test Suite.	GOTO: /link/to/ If the user has admin rights and fetches successfully: • Display the interface to adjust parameters for tests: ◦ Generative Model ◦ Mode ◦ Choose synthetic profiles ◦ Choose number of repetitions
2	Admin clicks "Save".	System persists the test suite document and displays a success message.